SUMAN DAS

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PROFESSIONAL SUMMARY

GenAI Architect and Agentic AI Lead with **15+ years of experience** building and scaling AI systems, now focused on **LLM-powered agentic platforms, RAG systems, and multi-agent orchestration** (LangGraph, CrewAI, MCP) in production. Known for **operating at platform and roadmap level**, translating emerging GenAI capabilities into **production-grade, reusable systems** adopted across teams. Deep multimodal foundation across vision-language models, fine-tuning (QLoRA), and enterprise ML. Regularly partners with enterprise architects and senior stakeholders to align AI initiatives with long-term business and technology strategy, with a demonstrated trajectory toward **broader organizational ownership beyond individual projects**.

WORK HISTORY

Lead AI Architect / GenAI Lead, 04/2022 - current
BOSCH GLOBAL SOFTWARE TECHNOLOGIES - Stuttgart, Bangalore

Agentic AI Platform & MCP Integration

- Architected an **agentic bug detection and auto-remediation pipeline** for microservices: LangGraph agents consume Azure Storage failure queues and Grafana/Loki logs, perform root-cause analysis against GitHub repositories, and raise fix PRs with human-in-the-loop review.
- Implemented a **cost-aware multi-LLM routing strategy** — frontier models (Claude, GPT-4) for code reasoning and repair, lightweight models (Gemini Flash) for high-volume log extraction and classification.
- Built **MCP (Model Context Protocol) servers and clients** exposing internal tools and data sources to agents in a standardized, reusable way; evangelized MCP adoption patterns across teams.
- Established **LLM evaluation and observability practices** (structured tracing, regression evals, guardrails) to move agentic workflows from demo to production reliability.

GenAI Multimodal Labeling & RAG Systems

- Designed a **GenAI-based attribute labeling approach** using Vision-Language Models to reduce dependency on manual annotation, positioned as a **reusable GenAI capability** beyond a single project.
- Built a **RAG-based similarity retrieval workflow** (embeddings + vector stores) with hybrid retrieval and reranking to improve labeling consistency at scale.
- Established **fine-tuning pipelines (QLoRA)** to adapt open-source VLMs and LLMs for domain-specific tasks, including structured-output prompting and context engineering at scale.

Agentic Workflow Orchestration - Autonomous Driving Data Platform

- Brought **agentic intelligence to existing event-driven data workflows** (video, LiDAR, radar processing), enabling higher throughput, self-healing behavior, and end-to-end transparency.
- Co-architected a **scalable, high-availability data platform** supporting large-scale model training; acted as **technical anchor** for ML teams, aligning platform capabilities with Bosch's long-term AI strategy.

Filter - Large-scale AI Training Data Generation

- Led high-throughput ML pipelines processing **terabytes of data daily**, scaling predictions from 50K to 400K images/day and delivering **1M+ labeled images** for downstream model development.
- Introduced CI/CD and infrastructure-as-code (Terraform) for deployment and model lifecycle management; **mentored ML engineers** and consulted across teams on architecture and GenAI decisions.

Senior Data Scientist, 04/2021 - 03/2022
ALSTOM TRANSPORT - Bangalore

Odometry

- Designed and owned an anomaly detection solution for train axle speed monitoring, achieving **99%+ recall**.
- Scaled from pilot to production across **1–100 trains**, automating maintenance alerts and eliminating manual inspection.

Senior Machine Learning Engineer, 06/2015 - 03/2021
TATA CONSULTANCY SERVICES LTD - Edinburgh, United Kingdom

Intelligent Document Extractor

- Led development of intelligent document extraction systems (NLP + CV) for UK banking clients, achieving **93% IoU** and production-scale deployment; evolved into **reusable products** adopted by multiple banks.

Churn Prediction

- Designed and deployed an XGBoost-based churn prediction model using behavioural and transactional data; implemented A/B testing to validate business impact prior to rollout.

Software Developer, 09/2009 - 01/2015
TATA CONSULTANCY SERVICES LTD - Kolkata

Backend Developer

- Designed and optimized data models, database schemas, and code for high-performance, scalable solutions; reduced maintenance costs by **£100K annually** through service and infrastructure improvements.

SKILLS

Core Expertise

- Agentic AI Systems & Multi-Agent Orchestration (LangGraph, CrewAI, MCP)
- Generative AI: LLMs, RAG (naive / hybrid / agentic), VLMs, Fine-tuning (QLoRA)
- LLM Evaluation, Observability & Production Reliability
- AI & Data Platform Architecture for Large-Scale Systems
- Technical Leadership, Architecture Alignment & Cross-Team Enablement

GenAI & Orchestration: LangGraph, CrewAI, MCP, RAG, Vector Databases (FAISS, pgvector), Prompt/Context Engineering, LLM Evals
Languages & ML Frameworks: Python, PyTorch, TensorFlow, XGBoost
Cloud & Infrastructure: Azure, AzureML, Kubernetes, Docker, Terraform, Ray
Data Systems: PostgreSQL, Databricks
Computer Vision: OpenCV, ResNet, U-Net, YOLO

EDUCATION

Master of Science: Physics, 06/2007 - 05/2009
Indian Institute of Technology - Roorkee - Honours

Bachelor of Science: Physics, 06/2004 - 05/2007
Calcutta University - Kolkata - Honours